

Protein-Protein Interaction Network Analysis for Endometriosis

Social Network Analysis course final project

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Which set of k proteins, when blocked simultaneously, most effectively limits the propagation of the disease signal in the endometriosis PPI network; and does the optimal set depend on the chosen measure?

Algorithm types used:

- group centrality measures,
- diffusion models.

Source:

- Genes from DisGeNET, UMLS C0014175, score ≥ 0.25 , 166 total
- Proteins from STRING, confidence ≥ 0.4 , first-shell expansion

Results:

- 454 nodes and 21,095 edges
- Single connected component
- Average degree ≈ 93 , density ≈ 0.205

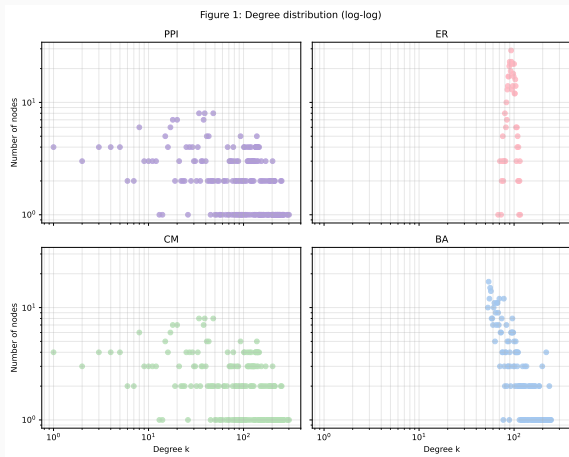
Random graphs as a sanity check: do the PPI numbers differ from random? All networks are matched on $V = 454$.

- Erdős-Rényi: random edges at 0.205 density
- Barabási-Albert: preferential attachment, $m \approx 53$
- Configuration model: preserved degree sequence

	$ E $	Density	Clustering	$\bar{\ell}$	Assortativity
PPI	21,095	0.205	0.637	1.94	0.052
ER	21,253	0.207	0.206	1.79	-0.005
BA	21,253	0.207	0.298	1.79	-0.019
CM	21,095	0.205	0.512	1.84	-0.181

PPI clustering and positive assortativity stand out

Degree distribution



PPI and CM heavy-tailed; ER narrow Poisson; BA power-law;

Simulate the disease signal spreading. Used to score blocking strategies by how much they reduce propagation.

- SIR, Susceptible-Infected-Recovered: infection rate β , recovery rate γ .
- IC, Independent Cascade: activation probability p .

$$\text{Spread reduction} = 1 - \frac{\text{spread for the blocked network}}{\text{spread for the unblocked network}}$$

Diffusion models' calibration

Calibrate β , γ , p so that unblocked propagation reaches around 70% of the network.

Parameters	Mean spread	% of network
SIR $\beta = 0.005, \gamma = 0.20$	318.7	70
SIR $\beta = 0.010, \gamma = 0.20$	382.3	84
SIR $\beta = 0.020, \gamma = 0.20$	418.6	92
SIR $\beta = 0.050, \gamma = 0.20$	439.5	97
IC $p = 0.005$	59.9	13
IC $p = 0.010$	170.5	38
IC $p = 0.020$	294.7	65
IC $p = 0.050$	386.0	85

Chosen: $\beta = 0.005, \gamma = 0.20, p = 0.02$.

Algorithms that pick proteins to block: two structural and one diffusion-aware.

- gdeg - greedy group degree: each step add the node whose neighborhood covers the most uncovered nodes
- gbtw - greedy iterative betweenness: each step remove the current highest-betweenness node
- CELF - Cost-Effective Lazy Forward: each step pick the node whose blocking best reduces simulated spread

CELF runs twice, once under SIR and once under IC.

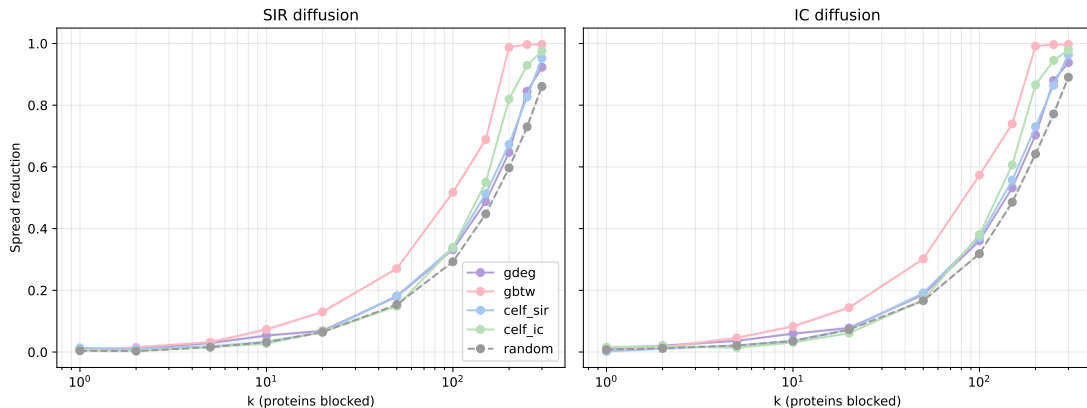
Top 10 picks per chain

Each chain's first ten proteins on the PPI graph.

	gdeg	gbtw	celf_sir	celf_ic
1	IL6	ALB	FGF4	CD40
2	ESR1	ESR1	KDR	LAG3
3	AKR1C3	EGFR	UGT1A7	CX3CL1
4	NCOA1	IL6	VTN	HDAC2
5	EGFR	GAPDH	IL17A	MED4
6	INS	TNF	NTRK3	UGT2B11
7	ITGB3	IL1B	PDGFRB	MED14
8	APOE	INS	SERPINF1	CCL13
9	CTNNB1	AKT1	EREG	PTPRC
10	CYP3A4	ACTB	TIMP1	COPS2

Spread reduction across strategies

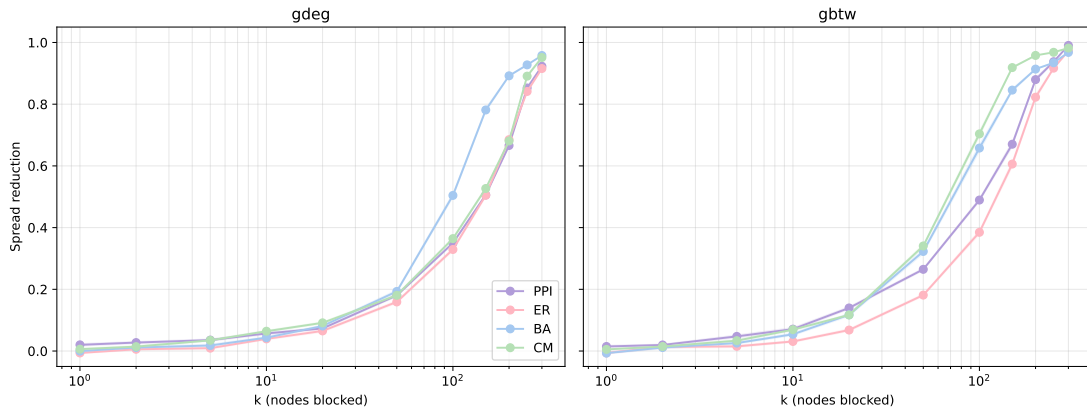
Figure 2: Spread reduction vs blocked-set size, shaded band shows SEM



gbtw dominates; saturates at $k \approx 200$ with 99% reduction.

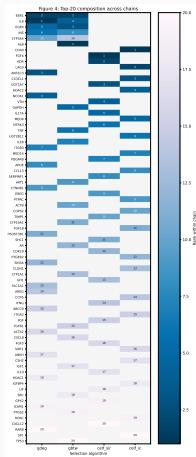
PPI vs null networks

Figure 3: Spread reduction on PPI vs null networks, shaded band shows SEM



PPI vs CM under gbtw: 7.8 pp gap - PPI is more resistant to blocking.

Top-20 composition



75 distinct proteins; only 5 shared by 2+ chains; 93% algorithm-specific.

Conclusion and limitations

The answer to the research question:

- Best k -set: prefix of the gbtw chain, saturates near $k = 200$
- Strongly algorithm-dependent: 93% of top-20 picks are algorithm-specific

At a realistic small k , partial blocking achieves a very modest reduction.

Main limitations: CELF inner replicates 25 not 50; CELF was not run on nulls; seed cutoff was fixed; single diffusion operating point.